



◀ SPANISH TERCIOS

Spanish royal troops fight in a deep pike formation during the Eighty Years' War (1568–1648) against the Dutch. The Spanish *tercios* were considered the best infantry in Europe.

KEY EVENTS

16TH–17TH CENTURY

- **1503** At Cerignola, the Spanish demonstrate the effectiveness of firearms and cannon deployed defensively behind field fortifications.
- **1513** Swiss pikemen attacking in dense columns beat the French at Novara.
- **1525** French armoured cavalry suffer heavy losses to Spanish arquebusiers in the defeat of François I by Habsburg forces at Pavia.
- **1580s** The matchlock musket comes into use.
- **c.1600** Dutch commander Maurice of Nassau trains and organizes his pike-and-musket infantry to carry out flexible battlefield manoeuvres.
- **1619–48** During the Thirty Years' War, the proportion of musket-armed infantry increases, sometimes outnumbering pikemen.
- **c.1660** The flintlock musket begins to replace the matchlock musket as the key European infantry weapon.

infantry was already apparent from the pike square – a man with a 5.5m (18ft) pike was formidable as part of a tight formation, but useless on his own. However, as the use of firearms increased, so did the need for further precision in drills: loading and firing the gun was a complex operation, particularly *en masse*, and its success in battlefield formation depended on tight discipline. The 17th century saw musket-armed troops manoeuvring in formation alongside pike squares, firing in volleys – either together for shock effect, or in sequence to maintain a continuous fire.

The impact of gunpowder weapons on European siege warfare, however, was far more rapid than that on field battles. Tall stone castle walls were obsolete by 1500, but during the Italian wars they were gradually replaced by star forts that were designed not only to resist bombardment by cannon, but also to enable the defenders to make optimal use of cannon and firearms themselves. As a result, sieges became even longer and more elaborate than before, with besieging infantry forced to spend long months in trenches that protected them from the defenders' firepower.



KEY STRUCTURE

STAR FORT

Originally called the “trace italienne”, the star fort's projecting triangular bastions enabled its defenders to cover all approaches to the walls with converging musket fire. The walls themselves were thick and low, surrounded by a wide ditch and an earth slope – the *glacis*.



▲ Plans for fortifications at Hunningen were drawn up in the 1670s. Star forts often had additional outworks, as here across the Rhine.